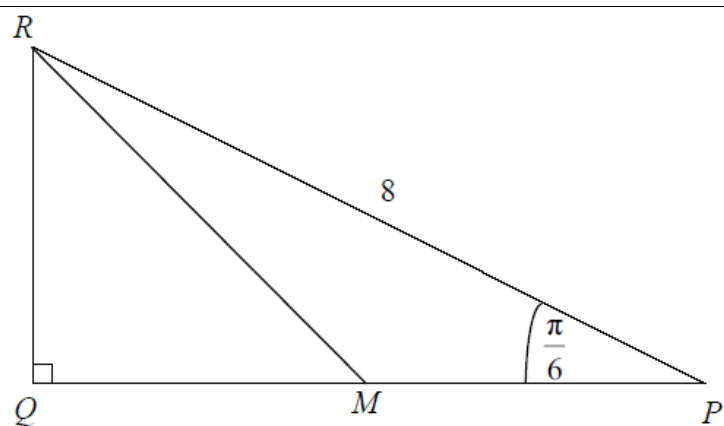


1	ACSB/2025/II/03
	(a) Prove that $\frac{1+\sin x}{\cos x} + \frac{\cos x}{1+\sin x} = 2 \sec x$. [4] (b) Hence solve the equation $\frac{1+\sin 2x}{\cos 2x} + \frac{\cos 2x}{1+\sin 2x} = -4$ for $0^\circ < x < 180^\circ$. [2]
2	ACSB/2025/II/05
	(a) State the amplitude and period of $y = 3 \cos\left(\frac{x}{2}\right) + 1$. [2] (b) Sketch the graph of $y = 3 \cos\left(\frac{x}{2}\right) + 1$ for $0 \leq x \leq 4\pi$. [3] (c) By drawing a suitable straight line on your sketch in part (b), determine the number of solutions of the equation $\cos\left(\frac{x}{2}\right) + 1 = \frac{x}{3\pi}$ for $0 \leq x \leq 4\pi$. [3]
3	BASS/2025/I/09
	(a) Find the amplitude and period of, (i) $-5 \cos 2x$, [2] (ii) $3 + 4 \sin x$. [2] (b) Sketch, on the same diagram, the curves $y = -5 \cos 2x$ and $y = 3 + 4 \sin x$ for $0 \leq x \leq 2\pi$ radians. [3]
4	CCHY/2025/I/9
	(i) Express $5 \sin^2 x - \cos^2 x - 1$ in the form $1 - k \cos 2x$. [2] (ii) State the amplitude and period, in radians, of $5 \sin^2 x - \cos^2 x - 1$. [2] (iii) Sketch the graph of $y = 5 \sin^2 x - \cos^2 x - 1$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. [3] (iv) By drawing the line $y = 1 - \frac{3x}{2\pi}$ on the same axes, state the number of solutions to [2] the equation $2\pi - 3x = 2\pi(5 \sin^2 x - \cos^2 x - 1)$ in the range $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.



The diagram shows a triangle PQR in which angle $RPQ = \frac{\pi}{6}$ radians.
 Angle RQP is a right angle, M is the mid-point of PQ and $PR = 8$ m.

Without using a calculator, find the value of the integer k such that
 angle $PRM = \sin^{-1}\left(\frac{\sqrt{k}}{14}\right)$.

[6]

The function f is defined by $f(x) = 4\cos 2x - 2$.

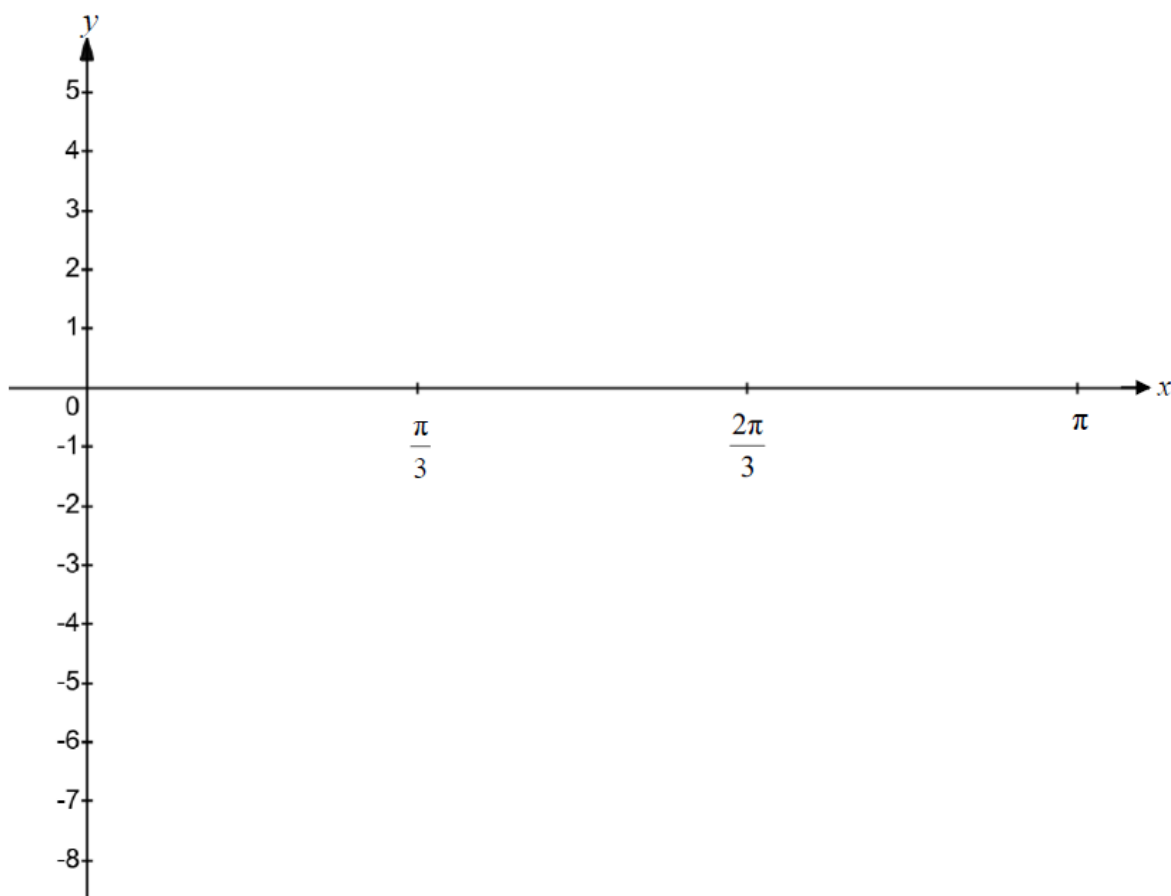
- (a) State the amplitude and period of f . [2]

The equation of a curve is $y = 4\cos 2x - 2$ for $0 \leq x \leq \pi$.

- (b) Find the coordinates of the minimum point of the curve. [2]

- (c) Find, in terms of π , the coordinates of the points where the curve meets the x -axis. [3]

- (d) Sketch the graph of $y = 4\cos 2x - 2$ for $0 \leq x \leq \pi$.



- (e) By drawing a suitable straight line on your sketch, determine the number of solutions of the equation $\cos 2x = \frac{x}{\pi} - 1$. [2]

1

$$\begin{aligned}\text{LHS} &= \frac{(1 + \sin x)^2 + \cos^2 x}{\cos x(1 + \sin x)} \\ &= \frac{1 + 2\sin x + \sin^2 x + \cos^2 x}{\cos x(1 + \sin x)}\end{aligned}$$

$$\begin{aligned}\text{(a)} \quad &= \frac{1 + 2\sin x + 1}{\cos x(1 + \sin x)} \\ &= \frac{2(1 + \sin x)}{\cos x(1 + \sin x)}\end{aligned}$$

$$\begin{aligned}&= \frac{2}{\cos x} \\ &= 2 \sec x \\ &= \text{RHS}\end{aligned}$$

$$2 \sec 2x = -4$$

$$\sec 2x = -2$$

$$\cos 2x = -\frac{1}{2}$$

(b)

$$2x = 120^\circ$$

$$x = 60^\circ$$

$$2x = 240^\circ$$

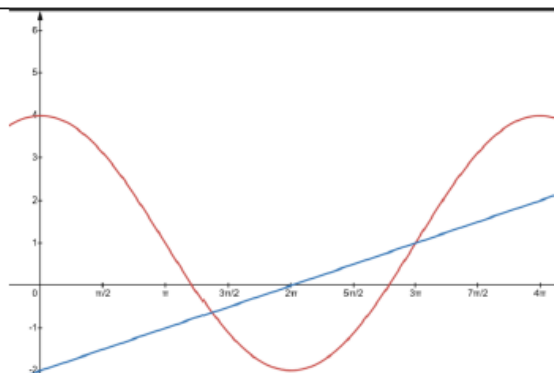
$$x = 120^\circ$$

2

(a) Amplitude = 3

Period = 4π

(b)



c)

$$3 \cos \frac{x}{2} + 3 = \frac{x}{\pi}$$

$$3 \cos \frac{x}{2} + 1 + 2 = \frac{x}{\pi}$$

$$y = \frac{x}{\pi} - 2$$

2 solutions

3

ai

Amplitude = 5

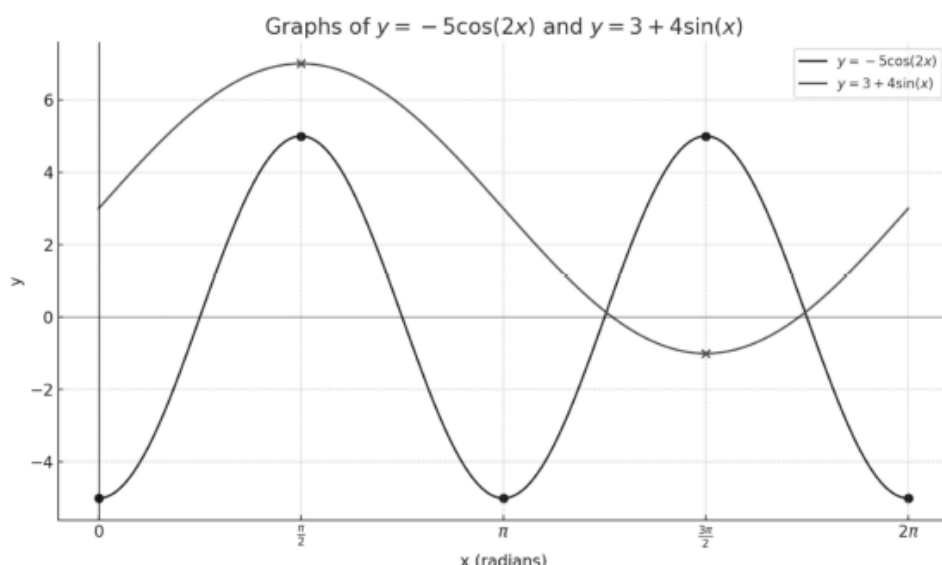
$$\text{Period} = \frac{2\pi}{2} = \pi \text{ or } 180^\circ$$

ai

Amplitude = 4

$$\text{Period} = 2\pi \text{ or } 360^\circ$$

b



4

(i)

$$\begin{aligned} 5\sin^2 x - \cos^2 x - 1 &= 5\left(\frac{1 - \cos 2x}{2}\right) - \left(\frac{\cos 2x + 1}{2}\right) - 1 \\ &= \frac{5}{2} - \frac{5\cos 2x}{2} - \frac{1}{2}\cos 2x - \frac{1}{2} - 1 \\ &= 1 - 3\cos 2x \end{aligned}$$

(ii)

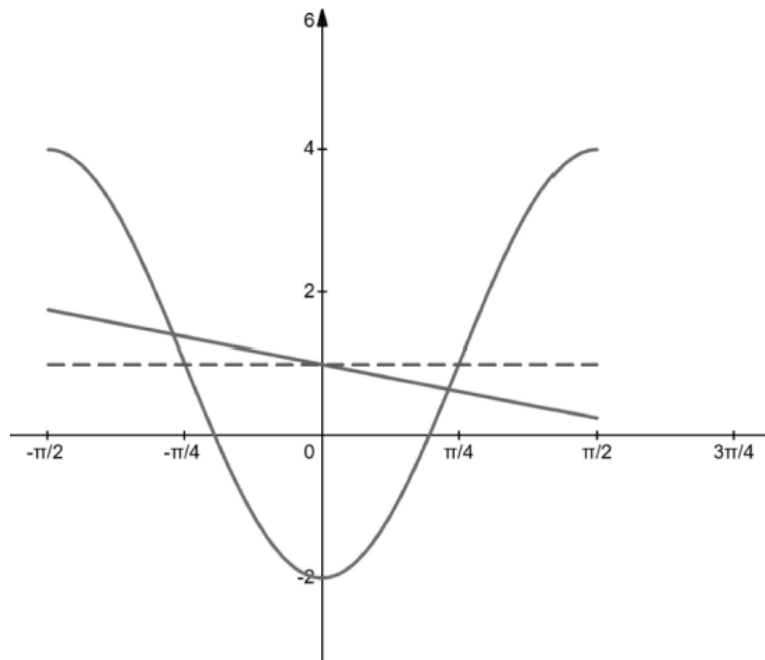
$$5\sin^2 x - \cos^2 x - 1 = 1 - 3\cos 2x$$

$$\text{Amplitude} = 3$$

$$\text{Period} = \frac{2\pi}{2}$$

$$= \pi \text{ radian}$$

(iii)



(iv)

$$y = -\frac{3}{2\pi}x + 1$$

$\therefore$ Number of solutions = 2.

5

$$\begin{aligned} \text{In } \triangle PQR, \quad \sin \frac{\pi}{6} &= \frac{QR}{8} \\ \frac{1}{2} &= \frac{QR}{8} \\ QR &= 4 \end{aligned}$$

$$\cos \frac{\pi}{6} = \frac{PQ}{8}$$

$$\frac{\sqrt{3}}{2} = \frac{PQ}{8}$$

$$PQ = 4\sqrt{3}$$

$$MP = MQ = 2\sqrt{3}$$

$$\begin{aligned} \text{In } \triangle QRM, \quad RM &= \sqrt{4^2 + (2\sqrt{3})^2} \\ &= \sqrt{28} \end{aligned}$$

$$\begin{aligned} \text{In } \triangle PRM, \quad \frac{\sin \angle PRM}{2\sqrt{3}} &= \frac{\sin \frac{\pi}{6}}{\sqrt{28}} \\ \sin \angle PRM &= \frac{\frac{1}{2}}{\sqrt{28}} (2\sqrt{3}) \end{aligned}$$

$$\begin{aligned}\sin \angle PRM &= \frac{\frac{1}{2}}{\sqrt{28}} (2\sqrt{3}) \\ &= \frac{\sqrt{3}}{\sqrt{28}} \\ &= \frac{\sqrt{3}}{\sqrt{28}} \left(\frac{\sqrt{28}}{\sqrt{28}} \right) \\ &= \frac{\sqrt{21}}{14} \\ \text{angle } PRM &= \sin^{-1} \left(\frac{\sqrt{21}}{14} \right) \\ k &= 21\end{aligned}$$

6

(a)

Amplitude = 4

Period = π

(b)

At minimum point, $y = -2 - 4$

$$= -6$$

$$x = \frac{\pi}{2}$$

Minimum point is $\left(\frac{\pi}{2}, -6 \right)$

(c)

$$4 \cos 2x - 2 = 0$$

$$\cos 2x = \frac{1}{2}$$

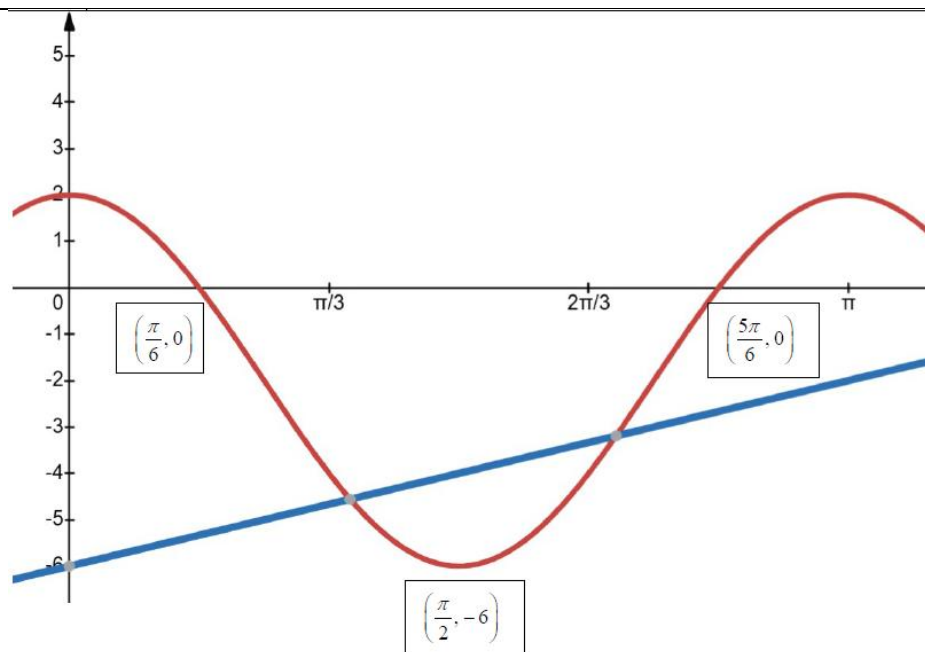
$$\text{Basic angle} = \frac{\pi}{3}$$

$$2x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

The points are $\left(\frac{\pi}{6}, 0 \right)$ and $\left(\frac{5\pi}{6}, 0 \right)$

(d)



(e)

$$\cos 2x = \frac{x}{\pi} - 1$$

$$4 \cos 2x = \frac{4x}{\pi} - 4$$

$$4 \cos 2x - 2 = \frac{4x}{\pi} - 6$$

Draw the line $y = \frac{4x}{\pi} - 6$

Number of solutions = 2